

CS2601 Linear and Convex Optimization

Homework 6

Due: 2023.12.12

1. Consider the equality constrained quadratic program

$$\begin{aligned} \min_{x_1, x_2} \quad & f(x_1, x_2) = x_1^2 + x_1 x_2 + x_2^2 - 2x_1 - 3x_2 \\ \text{s.t.} \quad & x_1 + 2x_2 = 1 \end{aligned}$$

- (a). Find the optimal solution x_1^*, x_2^* by reduction to an unconstrained problem.
- (b). Find the optimal solution and the corresponding Lagrange multiplier λ^* using the Lagrangian multipliers method.

2. Solve the following problem,

$$\begin{aligned} \min_{x_1, x_2} \quad & f(x_1, x_2) = x_1 x_2 \\ \text{s.t.} \quad & x_1^2 + 2x_2^2 = 1 \end{aligned}$$

3. In this problem, we characterize the eigenvalues of a symmetric matrix $\mathbf{A}$ as the optimal values of certain optimization problems. Let $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$ be the eigenvalues of $\mathbf{A}$.

- (a). Consider the following optimization problem,

$$\begin{aligned} \min_{\mathbf{x}} \quad & \mathbf{x}^T \mathbf{A} \mathbf{x} \\ \text{s.t.} \quad & \|\mathbf{x}\|_2^2 = 1 \end{aligned} \tag{1}$$

Use the Lagrange multipliers method to show that a solution $\mathbf{x}^*$ to problem (1) is an eigenvector of $\mathbf{A}$ associated to λ_1 , and the optimal value is λ_1 .

- (b). Let $\mathbf{v}_1$ be an eigenvector of $\mathbf{A}$ associated to the eigenvalue λ_1 . Consider the following optimization problem,

$$\begin{aligned} \min_{\mathbf{x}} \quad & \mathbf{x}^T \mathbf{A} \mathbf{x} \\ \text{s.t.} \quad & \|\mathbf{x}\|_2^2 = 1 \\ & \mathbf{v}_1^T \mathbf{x} = 0 \end{aligned} \tag{2}$$

Let $\mathbf{x}^*$ be an optimal solution to (2).

i) Use the Lagrange multipliers method to show that there exist some $c_0, c_1 \in \mathbb{R}$ s.t.

$$\mathbf{A}\mathbf{x}^* = c_0\mathbf{x}^* + c_1\mathbf{v}_1$$

ii) Show $c_1 = 0$ in i). Hint: Consider $\mathbf{v}_1^T \mathbf{A}\mathbf{x}^*$.

iii) Conclude $\mathbf{x}^*$ is an eigenvector of $\mathbf{A}$ associated to λ_2 , and the optimal value of (2) is λ_2 . You can assume the fact that an eigenvector orthogonal to $\mathbf{v}_1$ must be associated to one of the eigenvalues $\lambda_2, \dots, \lambda_n$.

Remark. The same argument can be used to show the following generalization. Let $\mathbf{v}_1, \dots, \mathbf{v}_{k-1}$ be linearly independent eigenvectors of $\mathbf{A}$ associated to $\lambda_1, \dots, \lambda_{k-1}$, respectively. Then λ_k is the optimal value of the following problem,

$$\begin{aligned} \min_{\mathbf{x}} \quad & \mathbf{x}^T \mathbf{A} \mathbf{x} \\ \text{s.t.} \quad & \|\mathbf{x}\|_2^2 = 1 \\ & \mathbf{v}_i^T \mathbf{x} = 0, \quad i = 1, 2, \dots, k-1 \end{aligned}$$

and the optimal solution is an eigenvector associated to λ_k .

4. Consider the following problem,

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^2} \quad & f(\mathbf{x}) = x_1^2 + x_2^2 \\ \text{s.t.} \quad & g_1(\mathbf{x}) = (x_1 + 1)^2 + (x_2 - 1)^2 - 1 \leq 0 \\ & g_2(\mathbf{x}) = x_1^2 + (x_2 - 1)^2 - 1 \leq 0 \end{aligned}$$

Write down the KKT conditions and find the optimal point $\mathbf{x}^*$ and the corresponding Lagrange multipliers.

5. Consider the following problem

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^2} \quad & f(\mathbf{x}) = x_1^2 + x_2^2 \\ \text{s.t.} \quad & (x_1 + 1)^2 + (x_2 - 1)^2 \leq 1 \\ & (x_1 + 1)^2 + (x_2 + 1)^2 \leq 1 \end{aligned}$$

(a). Sketch the feasible set and the level set of the objective function. Find the optimal point $\mathbf{x}^*$ and the optimal value f^* .

(b). Write down the KKT conditions. Do there exist Lagrange multipliers that satisfy the KKT conditions? Is $\mathbf{x}^*$ regular?

6. Consider the following problem

$$\begin{aligned} \min \quad & (x_1 - \frac{3}{2})^2 + (x_2 - \frac{3}{4})^2 \\ \text{s.t.} \quad & -x_1^2 + x_2 \geq 0 \end{aligned}$$

$$x_1 + x_2 \leq 4$$

$$x_1, x_2 \geq 0$$

For each of the following points, determine whether it is an optimal solution to the above problem and show your arguments,

$$\mathbf{x}^{(1)} = \begin{bmatrix} \frac{3}{2} \\ \frac{3}{4} \end{bmatrix}, \mathbf{x}^{(2)} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}, \mathbf{x}^{(3)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Hint: Try if you can find Lagrange multipliers satisfying the KKT conditions. Note you can easily check which constraints are active.